

NICHOLAS B. LUDWIG

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SF East Bay

SUMMARY

Systems engineer with six years production experience building agent harnesses, sandboxed Wasm runtimes, and developer platforms in Rust and Python. Led projects from prototype through production and maintenance. PhD researcher experienced in profiling, experimentation, and evidence-driven debugging.

EXPERIENCE

Independent Projects

May 2026 - Present

- Designed a Rust agent runtime, sieve, to resist prompt injection via sandboxing and planner isolation
- Various webapps to improve DevEx in a coding agent world, including Skin In The Game (repo owners can put a requirement that contributors must stake ETH to open PRs) and Prompt Request (for sharing agent conversation histories with humans or agents)
- Built agent evals spanning long-horizon WebAssembly-runtime implementation and scientific simulation/analysis; leading models consistently failed two and solved one intermittently
- Working on unnamed edtech product: an Aristotle-level tutor powered by learning science and LLMs

Hyperware (formerly Kinode/Uqbar)

February 2022 - May 2026

Lead Developer (promoted from Software Developer)

- Led core technical direction and owned roadmap, releases, infrastructure, and the 1.0 product launch
- Architected spider, an MCP agent harness with tool-call loops, persistent conversation, and orchestration across Hyperware process APIs; deployed to hundreds of personal servers
- Built hyperdrive, a Rust runtime for fault-tolerant, concurrent, sandboxed Wasm apps with message-passing and p2p communication; over 500 nodes deployed
- Developed document-driven AI coding workflow using Codex with custom async agent supervision
- Designed kit, CLI tooling to scaffold projects, run build/test/deploy workflows, and manage real and “fake” nodes (for development); adopted across Hyperware’s internal and community developer base
- Designed and open-sourced kibitz, an MCP coding agent with over 100 stars and 15 forks
- Shipped 10+ full-stack applications on Hyperware, from idea to working prototype in under a week, including a real-time multi-user p2p voice app over WebSocket
- Led and contributed to cross-functional app development: managed engineers extending DM-only p2p chat to CRDT-backed group chats; DAO portal for Hyperware smart contract interactions
- Redesigned OS homepage using React and iframes to deliver a polished mobile-OS-inspired experience
- Partnered with stakeholders to ship smart contracts within regulatory and technical constraints

Mod9 Technologies DBA Remeeting

May 2020 - February 2022

Backend Engineer

- Wrote Python REST API and asyncio SDK: simple, dev-friendly interfaces for ASR
- Improved maintainability of existing code and significantly reduced cost of infrastructure

University of Chicago

January 2015 - August 2019

Graduate Researcher

- Optimized simulator by trading memory for speed, reducing redundant computation: cached neighbors and lattice distance factors, used Ewald summation, and computed only the energy change each step

- Designed Python/Bash pipeline for running HPC jobs and analyzing results on Slurm: produced and post-processed tens to hundreds of CPU-years of compute
- Resolved ambiguous scientific questions via simulation; co-authored 4 papers including one in *Nature*

EDUCATION

University of Chicago

Ph.D., Physical Chemistry

December 2019

University of California, Merced

B.S., Chemical Sciences, with honors; Minor in Sociology

May 2014

SKILLS

Languages	Rust, Python, TypeScript, Go, C, Bash
AI/Agents	MCP, agent frameworks, Codex, Claude Code, tool use, LLM workflows
Systems/Data	perf, FlameGraph, Valgrind, Wasm, Tokio, Docker, WebSocket, OpenMP

SELECTED AGENT SYSTEMS

sieve	General-purpose agent framework designed to resist prompt injection
spider	MCP agent harness: tool loop, conversation storage, and Hyperware process orchestration
kibitz	Early coding agent (circa Jan 2025) for writing, testing, revising code via MCP tools